

**VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.08)**

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1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator V1.08. Version 1.08 adds a single-material design comparison mode that applies the previously validated minimum thickness design pathway across multiple candidate shielding materials.

The baseline attenuation model, NIST XCOM interpolation, point-source inverse-square model, isotope source model, Geometric Progression (G-P) buildup equation, and V1.07 target-driven minimum thickness calculations are unchanged. Therefore, this report focuses only on the new material comparison workflow layer.

V1.08 allows the simulator to answer a new design question: for the same source, detector distance, and target, how much of each available shielding material is required? The material comparison mode supports both manual monoenergetic photon sources and isotope sources. It returns one result candidate per selected material, preserves failed candidates instead of hiding them, keeps buildup fallback warnings, and sorts passing candidates by required shielding thickness.

This validation confirms that the V1.08 comparison layer correctly calls the V1.07 minimum thickness calculators, preserves pass/fail status, maintains warning behavior, and produces readable command-line output suitable for engineering comparison.

2. METHODOLOGY AND PHYSICS MODELS

2.1 Previously Validated Attenuation, Source, Buildup, and Target Models

V1.08 does not introduce a new attenuation equation, interpolation method, buildup factor equation, isotope source model, or minimum thickness solver. The underlying physics and target-driven design calculations remain based on the previously validated V1.07 implementation.

The narrow-beam attenuation model still calculates Beer-Lambert transmission using NIST XCOM mass attenuation coefficients converted to linear attenuation coefficients by material density. Detector flux is still calculated using isotropic point-source geometric spreading.

The isotope source model still processes photon lines line-by-line and sums source-level detector flux. The G-P buildup model remains limited to supported single-layer homogeneous shielding cases and is applied only within the valid mean-free-path range.

2.2 Material Comparison Workflow

The V1.08 material comparison workflow applies the same source, detector distance, target, maximum thickness setting, and buildup setting to multiple candidate materials. Each material is evaluated independently using the V1.07 minimum thickness calculator.

For manual monoenergetic sources, each candidate material is passed to the manual minimum thickness calculation pathway. For isotope sources, each candidate material is passed to the isotope minimum thickness calculation pathway. This lets V1.08 reuse the existing analytical and bisection-based design logic without modifying the underlying physics models.

The main comparison function returns a ranked list of candidates. Passing candidates are sorted from smallest required thickness to largest required thickness. Failed candidates are kept at the bottom of the result instead of being removed.

2.3 Candidate Result Model

Each material comparison entry is stored as a MaterialDesignCandidate. A candidate can either have status PASS or FAILED.

- PASS candidates store required thickness, final target-comparison flux, final effective transmission, buildup-used status, calculation mode, and warnings.
- FAILED candidates store the material and failure reason so the user can see why a material could not meet the requested target under the specified constraints.

The MaterialComparisonResult stores the target description, buildup setting, and complete candidate list. This separates the comparison data from the command-line output formatting and prepares the project for future optimization, plotting, and scenario-runner features.

2.4 Buildup and Fallback Behavior in Comparison Mode

When buildup is requested, V1.08 does not perform a new buildup calculation method. Instead, each candidate material uses the V1.07 buildup-aware minimum thickness design pathway. If buildup is valid, the candidate reports `buildup_used = True` and uses buildup-corrected flux as the target-comparison quantity.

If buildup is unsupported or unavailable within the valid G-P range, the candidate falls back to narrow-beam design and preserves the warning generated by the V1.07 minimum thickness layer. This allows unsupported buildup materials such as barite concrete and polyethylene to remain visible in material comparisons using narrow-beam fallback results.

2.5 command-line Interface

V1.08 adds a third command-line calculation mode: material comparison. The user can select a source, choose all materials or a comma-separated material list, enter detector distance, choose a target type, optionally set a maximum thickness, and choose whether to request G-P buildup.

The output table reports material name, required thickness, final flux, buildup status, and pass/fail status. Fallback warnings and failure reasons are printed below the table.

3. TEST CASE RESULTS

3.1 Material Comparison Structure and Target Satisfaction Tests

The validation runner checks that the V1.08 comparison layer returns one candidate per selected material and that passing candidates meet the requested flux target. The test source is Cs-137 at 1 Ci, detector distance 100 cm, FluxTarget(100 photons/cm²/s), and buildup enabled.

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	Cs-137 source, selected material list	One candidate per selected material	One candidate per selected material	PASS
2	Lead candidate	Status = PASS	Status = PASS	PASS
3	Lead candidate final flux	Final flux <= 100 photons/cm ² /s	Final flux <= 100 photons/cm ² /s	PASS
4	Lead candidate, buildup enabled	buildup_used = True	buildup_used = True	PASS
5	All PASS candidates	Final flux <= 100 photons/cm ² /s	Final flux <= 100 photons/cm ² /s	PASS

3.2 Buildup Fallback and Warning Preservation Tests

The validation runner verifies that unsupported buildup materials can still appear as passing narrow-beam fallback candidates when the target is reachable. The warning generated by the lower-level buildup-aware design pathway must be preserved in the comparison result.

Test Case	Input Parameters	Expected Result	Actual Result	Status
6	Concrete (Barite), buildup requested	Candidate status = PASS	Candidate status = PASS	PASS
7	Concrete (Barite), buildup requested	buildup_used = False	buildup_used = False	PASS
8	Concrete (Barite), buildup requested	Warning preserved	Warning preserved	PASS

3.3 Failed Candidate Preservation Tests

The comparison layer was also tested with an intentionally impossible target. The purpose of this test is to confirm that a failed material does not terminate the full comparison and does not disappear from the output.

Test Case	Input Parameters	Expected Result	Actual Result	Status
9	Cs-137, lead and water, FluxTarget($1 \cdot 10^{-30}$), max thickness = 0.1 cm	Lead candidate status = FAILED	Lead candidate status = FAILED	PASS
10	Failed lead candidate	Failure reason is stored	Failure reason is stored	PASS

3.4 Manual command-line Material Comparison Example

A manual command-line test was performed to confirm the V1.08 user-facing output is readable and useful for design comparison. The example compares all available materials for a Cs-137, 1 Ci source at 100 cm with a target flux of 100 photons/cm²/s and buildup requested.

Material	Required Thickness (cm)	Final Flux (photons/cm ² /s)	Buildup	Status
Tungsten	4.90817	$9.999991 \cdot 10^1$	Yes	PASS
Lead	7.08821	$9.999994 \cdot 10^1$	Yes	PASS
Copper	17.2589	$9.999997 \cdot 10^1$	Yes	PASS
Tin	18.6007	$9.999999 \cdot 10^1$	Yes	PASS
Concrete (Barite, Type BA)	28.7625	$9.999998 \cdot 10^1$	Fallback	PASS
Aluminum	58.3158	$9.999999 \cdot 10^1$	Yes	PASS
Concrete (Ordinary)	62.0647	$1.000000 \cdot 10^2$	Yes	PASS
Water	91.3324	$1.000000 \cdot 10^2$	Fallback	PASS
Polyethylene	94.6039	$1.000000 \cdot 10^2$	Fallback	PASS

The command-line output was sorted by required thickness, placed high-density shielding materials near the top of the table, and preserved fallback warnings for materials where buildup was unavailable or could not reach the target within the valid buildup range.

3.5 Regression Preservation

The validation runner still includes the previously validated regression and feature tests from earlier versions. These tests confirm that adding the material comparison layer did not alter the lower-level physics, source, buildup, or target-driven minimum thickness behavior.

Test Case	Input Parameters	Expected Result	Actual Result	Status
11	Manual source V1.05 lead shielding regression	Uncollided flux and buildup-corrected flux unchanged	Regression tests PASS	PASS
12	V1.06 isotope source calculations	Line counts, photon rates, source flux sums, and warnings remain valid	Source tests PASS	PASS
13	V1.07 minimum thickness design tests	Target conversion, bisection, buildup-aware design, and fallback behavior remain valid	Design tests PASS	PASS
14	Full validation runner	All validation tests pass	All validation tests passed	PASS

4. ASSUMPTIONS AND LIMITATIONS

1. Single-Material Solver Used for Each Candidate: V1.08 material comparison evaluates one material at a time by repeatedly applying the V1.07 single-material minimum-thickness solver.
2. Single Layer Homogeneous Buildup Only: G-P buildup correction remains supported only for single-layer homogeneous shielding cases.
3. Buildup-Aware Design is Single-Layer Only: Buildup-aware minimum thickness design is limited to single-layer homogeneous shielding because the implemented G-P buildup model is single-layer only.
4. No Multilayer Optimization: V1.08 does not optimize multilayer shielding stacks. Multilayer optimization is reserved for a later version.
5. No Multilayer Buildup Treatment: Multilayer buildup is not implemented. The simulator rejects buildup mode for multiple shielding layers because applying a single material buildup factor to mixed materials may produce misleading results.

6. Supported Buildup Materials: V1.08 supports G-P exposure buildup correction for lead, aluminum, ordinary concrete, water, tungsten, copper, and tin.
7. Unsupported Buildup Materials: Barite concrete and polyethylene remain unsupported for buildup correction in V1.08, although they remain available for narrow-beam attenuation calculations.
8. Point-Source Geometry: The source is still modeled as an isotropic point source.
9. Narrow-Beam Transmission Still Used as Baseline: The buildup-corrected flux is calculated by multiplying the narrow-beam uncollided flux by the G-P buildup factor.
10. Coefficient Interpolation: G-P coefficients are linearly interpolated between tabulated energies. This is separate from the NIST XCOM attenuation interpolation, which uses log-log interpolation.
11. Exposure Buildup Coefficients: The implemented coefficients are exposure buildup coefficients. Future versions may distinguish exposure, dose, and energy buildup factors if additional coefficient tables are added.
12. Simple 1D Geometry: The simulator assumes shielding thickness lies along the source to detector path. Therefore, total shielding thickness cannot exceed detector distance.
13. Selected Photon Lines: The isotope library introduced in V1.06 uses selected major photon lines, not complete emitted decay spectra.
14. Emitted Photons Before Source Self-Attenuation: Photon intensities are treated as emitted photons per decay before source encapsulation, source self-attenuation, air attenuation, or detector-window effects.
15. No Dose Conversion: V1.08 reports photon flux, not dose rate, exposure rate, air kerma, or effective dose.
16. G-P Range Limit: Buildup correction is only applied through 40 mean free paths. If a photon line exceeds this range, narrow-beam flux is reported but buildup is skipped with a warning.

5. CONCLUSION

The Shielding Attenuation Simulator V1.08 successfully adds single-material design comparison while preserving the previously validated attenuation, source, buildup, and target-driven minimum thickness models.

Validation confirmed that the material comparison workflow returns one candidate per selected material, identifies passing candidates, verifies that passing candidates meet the requested target flux, preserves buildup-used status, preserves fallback warnings, and keeps failed candidates visible with failure reasons.

Manual command-line testing confirmed that V1.08 produces a readable comparison table for a Cs-137 shielding design case. The table ranks materials by required thickness and clearly distinguishes buildup-aware results from narrow-beam fallback results.

V1.08 represents a practical improvement over V1.07 by allowing the simulator to answer a more useful engineering design question: for a given source and target, how much of each available shielding material is required? Future work should focus on constraint-based optimization using mass and cost, tradeoff plotting, scenario studies, case studies, CLI polish, GUI development, and eventual comparison against Monte Carlo transport tools.

References

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